Phase 3: External Building, Energy, Accessibility and Neighborhood Enrichment for Paris Apartment Valuation

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Abstract

This addendum evaluates authoritative external-data linkage for a Paris apartment automated valuation model. The 143,009-sale DVF Gold cohort is linked to BAN address identity, BDNB building characteristics, dated ADEME DPE records, Ile-de-France Mobilites stations, City of Paris schools and green spaces, BDNB shop/service equipment and 2022 strategic noise maps. Exact BAN matches cover 99.49% of sales; 99.997% receive a BDNB building and 76.85% receive a DPE that existed by the transaction date. No future-DPE joins are detected. A CatBoost model selected on 2024 and tested on 28,330 sales from 2025 achieves EUR 93,136 MAE, compared with EUR 96,428 for the Phase 2 correction. The paired MAE reduction is EUR 3,293 (95% bootstrap interval: EUR 2,735–3,846). Static source snapshots are explicitly treated as retrospective reconstruction, not fully vintage-correct historical features.

1 Data integration

The Bronze layer stores immutable source snapshots and a manifest containing URLs, producer, licence, retrieval time, byte size and SHA-256. The Silver layer records row-level entity resolution. BAN is matched deterministically by commune, FANTOIR road code, number and suffix. Official BAN–BDNB relations are ranked by reliability; the cadastral parcel is a fallback. The Gold table has one row per original sale and 141 columns.

Building features include construction year/period, level count, height, footprint, dwelling/lot counts, wall/roof fields and coarse state. Location features include distance/density for Metro, RER, train, tram, schools, public green spaces and BDNB equipment, plus road/RATP/SNCF Lden bands.

2 Temporal rules

For a sale at date t, an ADEME diagnostic is eligible only when

$$\texttt{date\_etablissement\_dpe} \leq t.$$

The historical comparable layer retains Phase 2's 90-day availability lag. BAN, most BDNB physical attributes, strategic noise and amenity/transport snapshots are labelled retrospective static context because consistent annual vintages are unavailable. Ablations separate dated DPE from static building and context information.

3 Results

The selected CatBoost model improves MAE by EUR 3,293 (3.42%). Its paired 95% bootstrap interval, EUR 2,735–3,846, excludes zero. Dated DPE alone improves MAE by EUR 1,206; building-only and context-only ablations improve by EUR 1,858 and EUR 796 respectively. The high-cardinality identity layer adds a further gain relative to full LightGBM without raw identity.

Table 1: Frozen 2025 chronological test.

Method	MAE (EUR)	MdAPE	Within 20%	R^2
CatBoost, full identity/enrichment	**93,136**	**11.70%**	**73.39%**	**0.8679**
LightGBM, full without identity	93,968	11.80%	73.03%	0.8661
LightGBM, building only	94,571	11.90%	72.28%	0.8658
LightGBM, dated DPE only	95,222	11.97%	72.28%	0.8637
LightGBM, context only	95,632	11.99%	71.74%	0.8627
Phase 2 correction	96,428	12.22%	71.69%	0.8601

4 Limitations and deployment

Public sources did not expose a reliable apartment-floor or elevator field. Both are stored as explicitly unavailable rather than inferred from building height or level count. Static 2026 building/context reconstruction may improve a retrospective 2025 backtest but is not proof of information availability in 2021–2025. A prospective 2026 or later cohort and vintage-correct source snapshots are required before a strong external deployment claim. The model is an automated indication, not a certified appraisal.

5 Reproducibility

The source manifest, match audit, feature-quality contract, validation trials, row-level test predictions, paired intervals and selected model artifact are saved with the project. The pipeline is run in order with `acquire_phase3_data.py`, `build_phase3_features.py` and `benchmark_phase3.py`.